Golden_Kelly_Submission_1

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Resources Used: Claude (Anthropic) was used to help me determine which logical path made the most sense and navigate my code errors.

Variables Chosen:

I'm looking at CXCR2, a receptor on neutrophils that responds to the IL-8 inflammatory signal. Inflammation is thought to play a role in preeclampsia, which is what made CXCR2 an interesting gene to explore. My continuous covariate is maternal vitamin D (mat_vitD), measured as 25-hydroxyvitamin D in ng/mL at baseline.

The two categorical covariates are pregnancy outcome (preg_preeclampsia: normal vs. preeclampsia) and obesity status. I created obesity status from maternal BMI (mat_BMI), using the WHO cut point: anyone with a BMI of 30 kg/m^2 or higher is classified as obese.

Upload the data and clean a little of it up

```
#read_csv() automatically strips the invisible character (BOM) at the start
#of the covariate file, so the first column is named "subjectID" correctly.
covar <- read_csv("/Users/kellygolden/Library/CloudStorage/OneDrive-DartmouthCollege/HSE 103/Final Proj

#gene expression: rows = probe IDs, columns = samples (GEO IDs).
#the first column has no name in the file, so rename it PROBEID.
geneExp <- read_csv("/Users/kellygolden/Library/CloudStorage/OneDrive-DartmouthCollege/HSE 103/Final Pr
  rename(PROBEID = 1)

#linker file: probe ID -> gene name
geneNames <- read_csv("/Users/kellygolden/Library/CloudStorage/OneDrive-DartmouthCollege/HSE 103/Final

dim(covar)
```

```
## [1] 157  10
```

```
dim(geneExp)
```

```
## [1] 500 158
```

Cleaning! Each covariate value (except the IDs) has a text label in front of it, like `"age_years: 31...."` or `"pregnancy_condition: Normal"`. I have to remove everything up to and including the `": "` and then convert to the right type of variable.

```
#Look at the raw format first
head(covar$mat_age_yrs, 3)
```

```
## [1] "age_years: 31.06639288" "age_years: 27.79466119" "age_years: 34.31895962"
```

```
#columns 3 to 10 have the "label: value"
for (col in names(covar)[3:10]) {
  covar[[col]] <- sub(".*: ", "", covar[[col]])   #keep only the part after ": "
```

```
}#Claude helped with this cleaning code

#convert from text to numbers
numVars <- c("mat_age_yrs", "delivery_gestAge_wks",
             "sampleCollection_gestAge_wks", "mat_BMI", "mat_vitD")
for (col in numVars) {
  covar[[col]] <- as.numeric(covar[[col]])
}

#convert categorical variables to factors with clear labels (Claude helped here)
covar$preg_preeclampsia <- factor(covar$preg_preeclampsia,
                                  levels = c("Normal", "Preeclampsia"))
covar$mat_asthma <- factor(covar$mat_asthma, levels = c(0, 1),
                           labels = c("Healthy", "Asthma"))
covar$batch_Microarray <- factor(covar$batch_Microarray)

#derived categorical variable: obesity status from BMI (WHO: BMI >= 30)
covar$obesity <- factor(ifelse(covar$mat_BMI >= 30, "Obese", "Not obese"),
                        levels = c("Not obese", "Obese"))

#confirm no missing values were created by the conversions (Claude helped here)
colSums(is.na(covar))
```

```
##                    subjectID                        geoID
##                            0                            0
##                  mat_age_yrs           preg_preeclampsia
##                            0                            0
##         delivery_gestAge_wks sampleCollection_gestAge_wks
##                            0                            0
##             batch_Microarray                      mat_BMI
##                            0                            0
##                      mat_vitD                   mat_asthma
##                            0                            0
##                      obesity
##                            0
```

```
summary(covar[, c(numVars, "preg_preeclampsia", "obesity")])
```

```
##   mat_age_yrs    delivery_gestAge_wks sampleCollection_gestAge_wks
##  Min.   :18.43   Min.   :26.43        Min.   :10.00
##  1st Qu.:22.15   1st Qu.:38.14        1st Qu.:12.14
##  Median :26.57   Median :39.29        Median :13.43
##  Mean   :26.38   Mean   :38.82        Mean   :14.13
##  3rd Qu.:30.14   3rd Qu.:40.00        3rd Qu.:16.43
##  Max.   :38.76   Max.   :42.00        Max.   :18.86
##     mat_BMI         mat_vitD        preg_preeclampsia      obesity
##  Min.   :17.95   Min.   : 5.11   Normal      :110     Not obese:100
##  1st Qu.:22.90   1st Qu.:13.40   Preeclampsia: 47     Obese    : 57
##  Median :27.33   Median :20.40
##  Mean   :28.94   Mean   :22.98
##  3rd Qu.:33.27   3rd Qu.:29.50
##  Max.   :61.95   Max.   :80.80
```

Match the probes to the ID names

```
#Claude helped me what functions to apply for this section to get to the desired
#result

#confirm every probe in the expression data has a gene name in the linker file
stopifnot(all(geneExp$PROBEID %in% geneNames$PROBEID))

geneExp <- geneExp %>%
  left_join(geneNames, by = "PROBEID") %>%   #add a GENE column
  relocate(GENE, .after = PROBEID)           #move it next to PROBEID

#ensure no gene name appears twice (each gene is one probe)
stopifnot(!any(duplicated(geneExp$GENE)))
```

Merge: The expression data has samples as columns, while the covariates have samples as rows. Reshaped the expression data to long format (one row per gene per sample), kept CXCR2, and joined it to the covariates by geoID.

```
gene <- "CXCR2"

geneLong <- geneExp %>%
  select(-PROBEID) %>%
  pivot_longer(cols = -GENE, names_to = "geoID", values_to = "expression")

dat <- geneLong %>%
  filter(GENE == gene) %>%
  inner_join(covar, by = "geoID")

#verify same samples in both files, none lost or duplicated (Claude helped here)
stopifnot(setequal(covar$geoID, unique(geneLong$geoID)))
stopifnot(nrow(dat) == nrow(covar), !any(duplicated(dat$geoID)))

#checking one sample to confirm the joined value=raw value (Claude helped here)
check_id <- dat$geoID[1]
raw_value <- geneExp %>% filter(GENE == gene) %>% pull(all_of(check_id))
stopifnot(dat$expression[1] == raw_value)

nrow(dat)
```

```
## [1] 157
```

Plot theme!

```
theme_pub <- theme_classic(base_size = 12) +
  theme(axis.title   = element_text(face = "bold"),
        axis.text    = element_text(color = "black"),
        legend.title = element_text(face = "bold"),
        legend.position = "top")

obesityColors <- c("Not obese" = "#56B4E9", "Obese" = "#E69F00")
```

Plot 1: Histogram of CXCR2 expression

```
ggplot(dat, aes(x = expression)) +
  geom_histogram(bins = 25, fill = "#0072B2", color = "white") +
  labs(x = "CXCR2 expression (log2 normalized intensity)",
       y = "Number of participants") +
```

```
  scale_y_continuous(expand = expansion(mult = c(0, 0.05))) +  #bars sit on the axis
  theme_pub
```

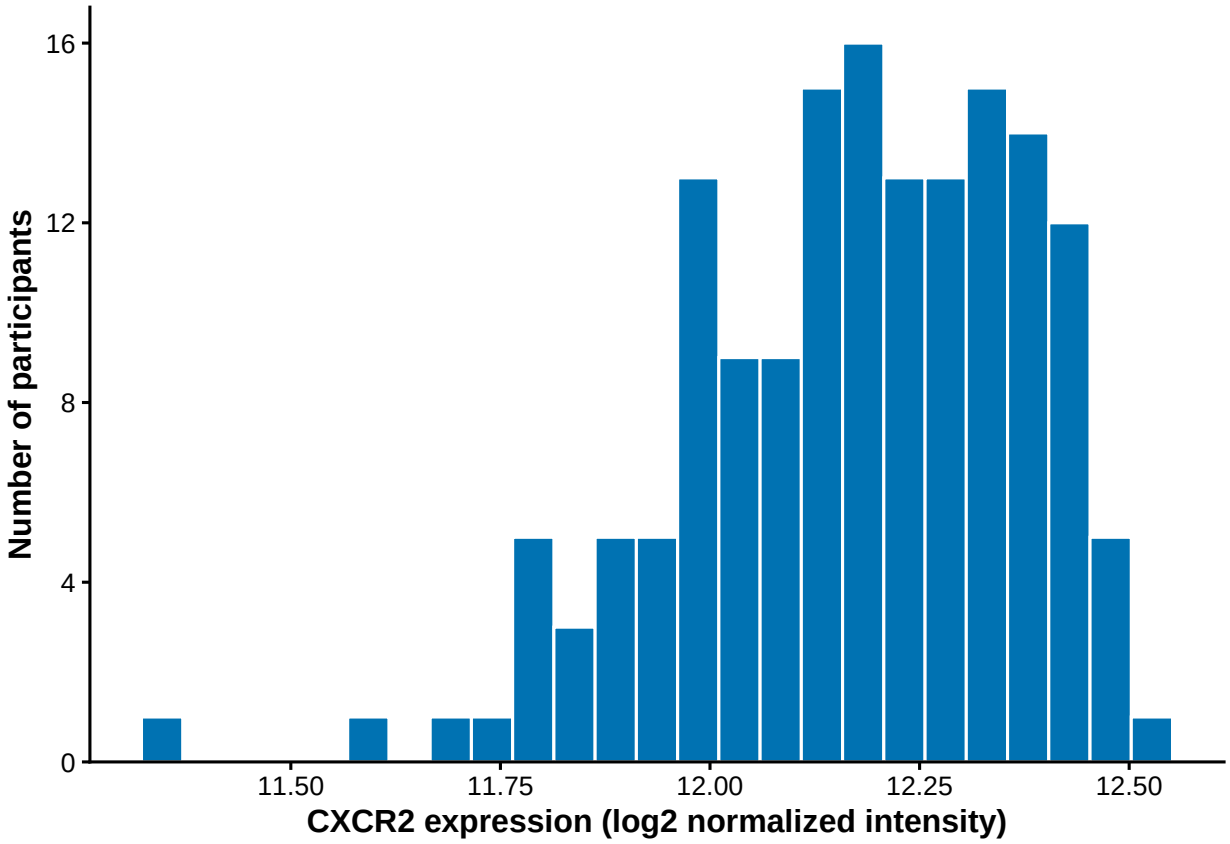


Figure 1: **Distribution of CXCR2 expression.** Expression values from 157 pregnant participants are log2-transformed normalized intensities from the Affymetrix Human Gene 1.0 ST Array, measured in blood collected at 10–19 weeks' gestation.

Plot 2: Scatterplot of vitamin D vs. CXCR2 expression

```
ggplot(dat, aes(x = mat_vitD, y = expression)) +
  geom_point(color = "#0072B2", alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "black", linewidth = 0.8) +
  labs(x = "Maternal 25-hydroxyvitamin D (ng/mL)",
       y = "CXCR2 expression (log2)") +
  theme_pub
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Plot 3: Boxplot of CXCR2 by pregnancy outcome and obesity status

```
ggplot(dat, aes(x = preg_preeclampsia, y = expression, fill = obesity)) +
  geom_boxplot(outlier.shape = NA, width = 0.6,
               position = position_dodge(width = 0.7)) +
  #show the participants on top of the boxes
  geom_point(position = position_jitterdodge(jitter.width = 0.15,
                                             dodge.width = 0.7),
             size = 1.2, alpha = 0.5) +
  scale_fill_manual(values = obesityColors, name = "Obesity status") +
```

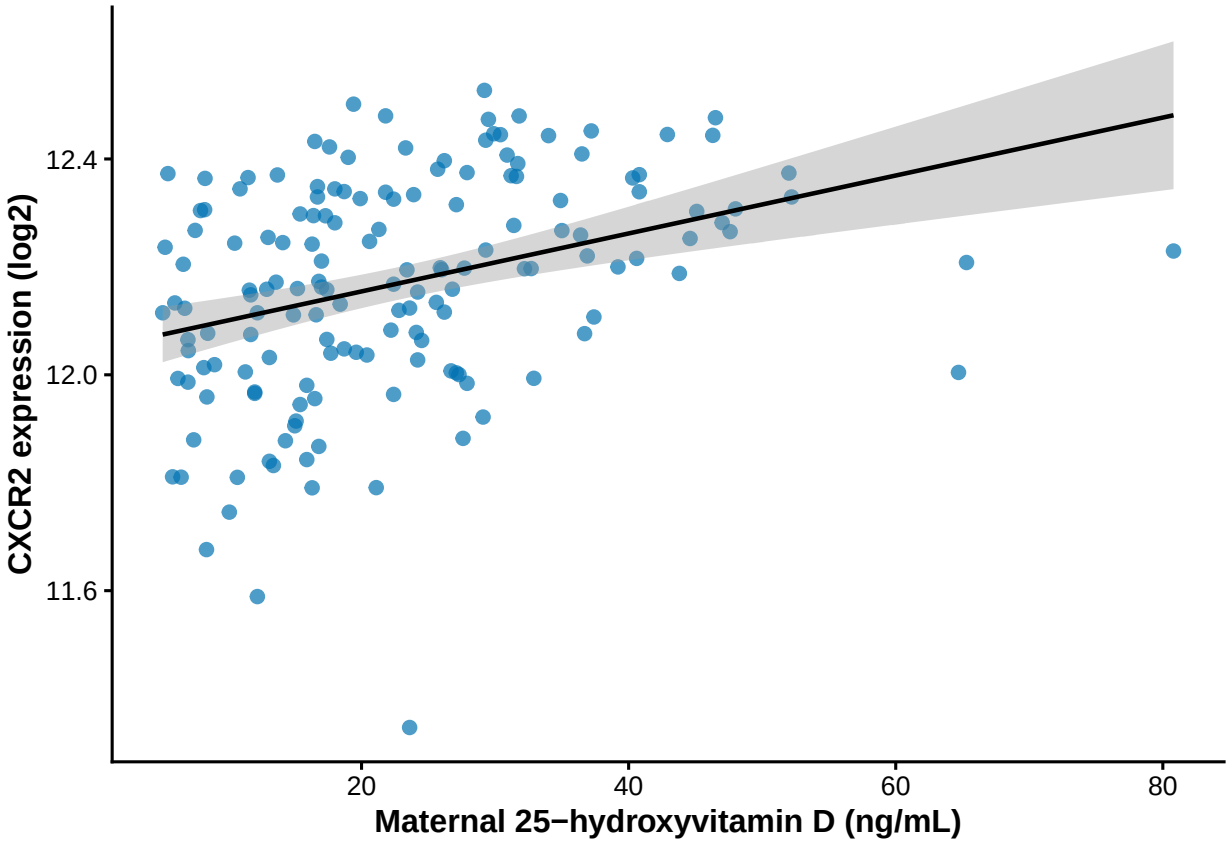


Figure 2: **Maternal vitamin D vs. CXCR2 expression.** Each point is one participant (n = 157). The black line is a linear regression fit, and the gray band is its 95% confidence interval.

```
  labs(x = "Pregnancy outcome", y = "CXCR2 expression (log2)") +
  theme_pub
```

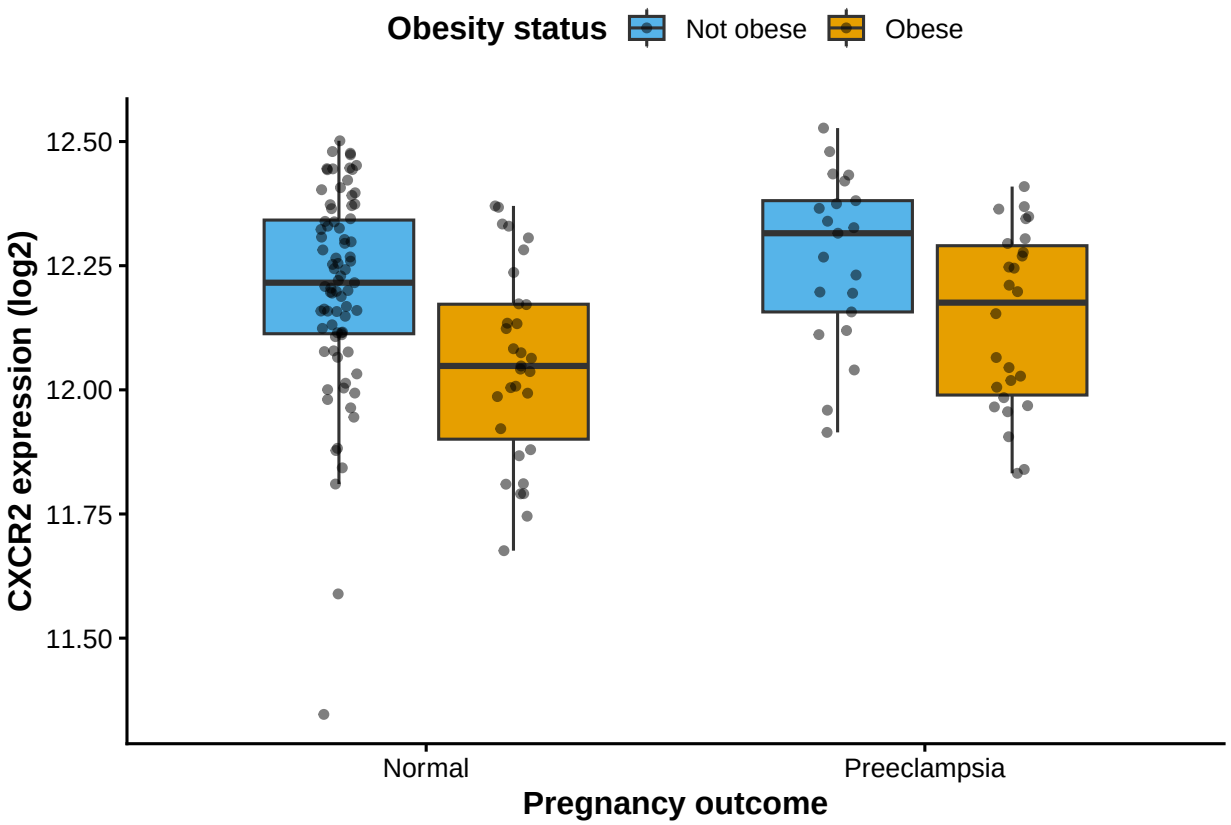


Figure 3: **CXCR2 expression by pregnancy outcome and obesity status.** Obesity was defined as BMI 30 kg/m². Boxes show the median and interquartile range, whiskers extend to 1.5 × IQR, and points are individual participants. Group sizes: Normal/not obese n = 79, Normal/obese n = 31, Preeclampsia/not obese n = 21, Preeclampsia/obese n = 26.